



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Advanced)
UNIT TEST
17-04-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : TOAS]

Time : 3 Hours

Maximum Marks : 186

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **32** pages and that all the **18** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. A solid non-conducting hemisphere of radius R has a uniformly distributed positive charge of volume density ρ . A negatively charged particle having charge q is transferred from centre of its base to infinity. Work performed in the process is :

(A) $\frac{q\rho R^2}{4\epsilon_0}$
(B) $\frac{q\rho R^2}{2\epsilon_0}$
(C) $\frac{q\rho R^2}{8\epsilon_0}$
(D) $\frac{-q\rho R^2}{8\epsilon_0}$
2. The potential of electrostatic field is given by $\phi = xy$, where x & y are coordinate. Which of the following is correct ?

(A) Electric field at all points on positive x -axis is in positive y -direction.

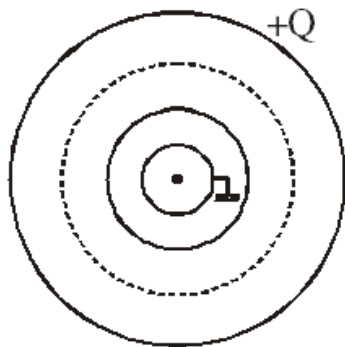
(B) Electric field at all points of positive y -axis is in negative x -direction.

(C) Electric field at $(2, 2)$ is $\sqrt{2}\text{N/m}$.

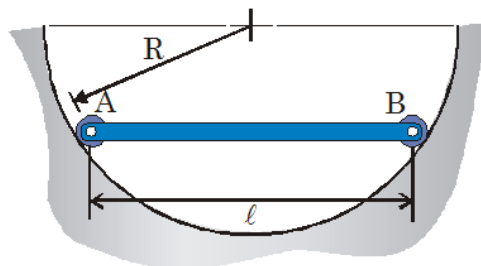
(D) Electric field at x, y makes an angle $\theta = \tan^{-1} \left(\frac{y}{x} \right)$ with x -axis.

Space for Rough Work

3. Consider a system of n concentric conducting shells of radii $r, 2r, 4r, 8r, \dots, 2^{n-1}r$ respectively. If inner shell is earthed and Q charge is given on outer most shell, charge appears on inner most shell is :-



- (A) $-Q$ (B) $-\frac{Q}{2^n}$ (C) $-\frac{Q}{2^{n-1}}$ (D) $+Q$
4. The bar has a length ℓ and mass m . It is supported at its ends by rollers of negligible mass. If it is given a small displacement and released, determine the natural frequency of vibration.



- (A) $\omega_N = \sqrt{\frac{3g(4R^2 - \ell^2)}{6R^2 - \ell^2}}$ (B) $\omega_N = \sqrt{\frac{3g(4R^2 - \ell^2)^{\frac{1}{2}}}{6R^2 + \ell^2}}$
- (C) $\omega_N = \sqrt{\frac{3g(4R^2 - \ell^2)^{\frac{1}{2}}}{6R^2 - \ell^2}}$ (D) $\omega_N = \sqrt{\frac{3g(4R^2 + \ell^2)^{\frac{1}{2}}}{6R^2 - \ell^2}}$

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

5. Which of the following relations for a coaxial lens-system separating two media of refractive indices μ_1 and μ_2 are true ? (Here u and v are the distances of the object and the image from the first and the second principal points respectively, f_1 and f_2 the focal lengths of the system and m is linear transverse magnification.)

(A) $\frac{f_1}{u} + \frac{f_2}{v} = 1$

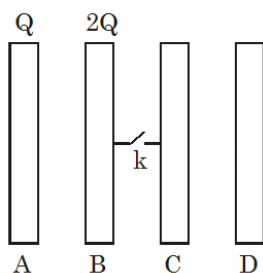
(B) $m = -\frac{v}{u} \left(\frac{f_1}{f_2} \right)$

(C) $\frac{f_1}{f_2} = -\frac{\mu_1}{\mu_2}$

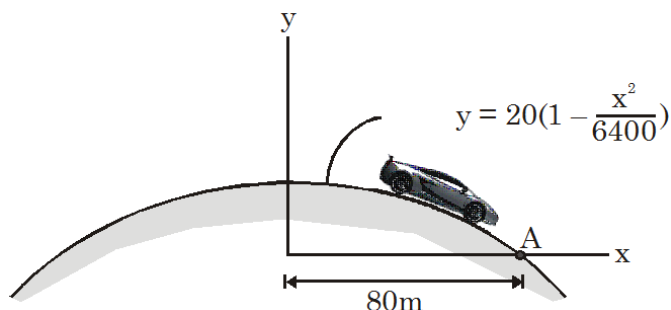
(D) $\frac{f_1}{u} - \frac{f_2}{v} = 1$

Space for Rough Work

6. Initially two plates (A and B) are given charges as shown. Find the final charges on the plates after the key k is closed. All plates are conducting, parallel and of infinite length and breadth.



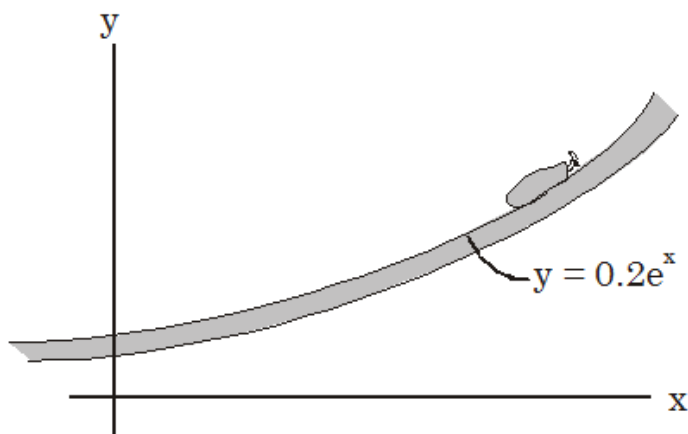
- (A) Final charges on plate A is Q (B) Final charge on plate B is 0
 (C) Final charge on plate C is $3Q/2$ (D) Final charge on plate D is 0
7. The 0.8-Mg car travels over the hill having the shape of a parabola. When the car is at point A, it is traveling at 9 m/s and increasing its speed at 3 m/s^2 . Choose the correct option(s): (Neglect the size of the car)



- (A) The resultant normal force is 6.86 kN
 (B) The resultant normal force is 2.73 kN
 (C) Resultant frictional force that all the wheels of the car exert on the road at this instant is 2.22 kN.
 (D) Resultant frictional force that all the wheels of the car exert on the road at this instant is 1.18 kN.

Space for Rough Work

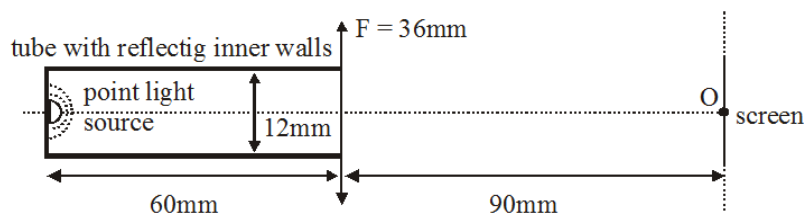
8. The 8-kg sack slides down the smooth ramp. If it has a speed of 1.5 m/s when $y = 0.2$ m, choose the correct option(s) :



- (A) Normal reaction the ramp exerts on the sack is 78.44 N
 (B) Normal reaction the ramp exerts on the sack is 81.84 N
 (C) The rate of increase in the speed of sack at this instant is 1.96 m/s^2
 (D) The rate of increase in the speed of sack at this instant is 3.92 m/s^2

Space for Rough Work

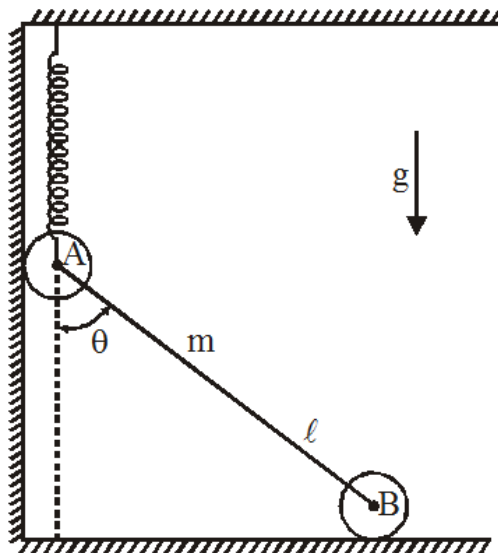
9. A tube with reflecting inner walls has a point light source at the bottom of it (see figure). The inner diameter of the tube is $d = 12 \text{ mm}$, the tube's length is $\ell = 60 \text{ mm}$. A convex lens with a focal length $F = 36 \text{ mm}$ is placed against the open end of the tube and at a distance $L = 90 \text{ mm}$ from the tip of the tube there is a screen, which has a graph paper attached to it. The intersection point with the optical axis O is marked on the paper.



- (A) A point image is formed on the screen at point O.
 (B) The pattern of images obtained on the screen are parallel straight lines.
 (C) The pattern of images obtained on the screen are concentric circles.
 (D) The radius of concentric circles formed on the screen are $(18n) \text{ mm}$, where n is an integer.

Space for Rough Work

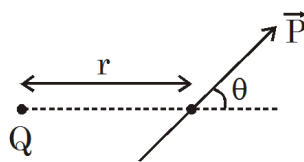
10. Two identical discs, each of negligible mass and radius $r \ll \ell$ are connected by a uniform rod of mass m and length ℓ as shown in the figure. The disc A is connected with an ideal spring of spring constant k at its centre. The spring is unstretched when $\theta = 0^\circ$. There is no slipping at any contact at any point of time during the motion. Choose the correct statement(s) [Spring always obeys Hook's Law and $mg = k\ell$]. During the motion, each disc remains in contact with the contact surface :-



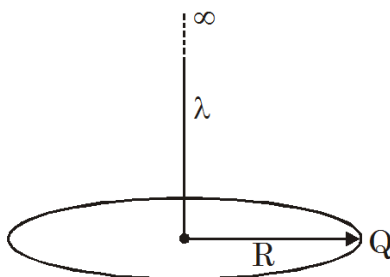
- (A) The value of θ , when the system is in stable equilibrium is 60°
- (B) The value of θ , when the system is in stable equilibrium is 30°
- (C) Spring potential energy of the system in the state of stable equilibrium is $\frac{k\ell^2}{8}$
- (D) The angular speed of the rod when it passes through the position of the stable equilibrium is $\sqrt{\frac{3g}{4\ell}}$

Space for Rough Work

11. A dipole of dipole moment $\vec{P}$ is placed in front of a point charge as shown. Choose the correct statement(s) :



- (A) The dipole will experience a net force irrespective of value of θ .
 (B) The dipole will experience a net torque about its center irrespective of value of θ .
 (C) Force experience by dipole is given by $\frac{kQP}{r^3} \sqrt{1 + 3\cos^2\theta}$.
 (D) If charge Q is not fixed, the whole system can't be in equilibrium for any value of ' r ' & θ .
12. Diagram shows a semi-infinite wire having a charge density λ . A ring carrying a total charge Q is kept as shown. Choose the correct option(s) :



- (A) The force experienced by ring is in vertical direction & have a value of $\frac{\lambda Q}{4\pi\epsilon_0 R}$
 (B) Net force experienced by ring is in vertical direction and have value of $\frac{\lambda Q}{2\pi\epsilon_0 R}$
 (C) Tension in the ring is zero.
 (D) Tension in the ring is $\frac{Q\lambda}{8\pi^2\epsilon_0 R}$

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

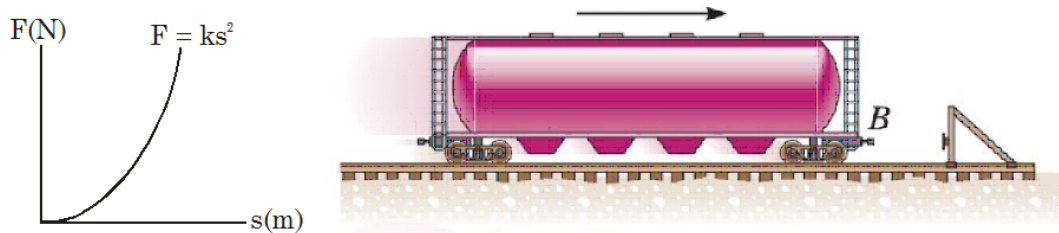
Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

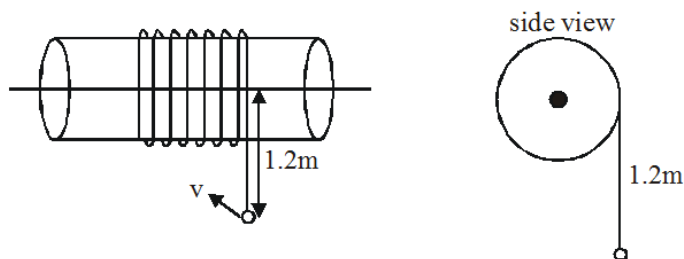
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1. Two point light sources are positioned on different points of a convex lens' optical axis. The images of these light sources formed by the lens coincide. It is known that one light source is at a distance $a = 18$ cm from the center of the lens. How far (in cm) is the other light source distanced from this light source? The focal length of the lens is $f = 9$ cm.
 2. Find the coulomb attraction (in N) on a charge of magnitude 5×10^{-4} C at the origin due to a charge (of constant density ρ) occupying the volume inside the cone $z^2 = x^2 + y^2$, $0 < z < h$. (Take $\rho = \frac{\sqrt{2}}{\pi} \mu\text{C}/\text{m}^3$, $h = (\sqrt{2} + 1)$ m)
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Space for Rough Work

3. Design considerations for the bumper B on the 5000 kg train car require use of a nonlinear spring having the load-deflection characteristics shown in the graph. Select the proper value of k (in MN/m^2) so that the maximum deflection of the spring is limited to 0.2 m when the car, traveling at 4 m/s, strikes the rigid stop. Neglect the mass of the car wheels.

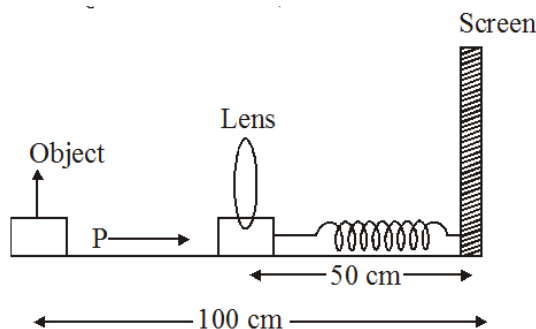


4. A string is wrapped over a fixed horizontal cylinder of radius $\frac{7}{66}$ m and its other end is connected to a point mass, which hangs in air such that 1.2 m length of string is free. Find minimum velocity (in m/s) given to mass perpendicular to length of cylinder, so that string never gets slack before wrapping over cylinder completely once. (Take $g = 9.8 \text{ m/s}^2$)

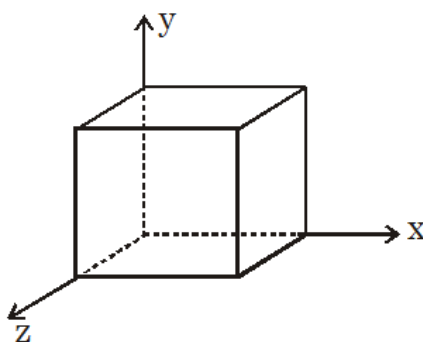


Space for Rough Work

5. A point object is located at a distance of 100 cm from a screen. A lens of focal length 23 cm mounted on a movable frictionless stand is kept between the source and the screen. The stand is attached to a spring of natural length 50 cm and spring constant 800 N/m as shown. Mass of the stand with lens is 2 kg. How much minimum impulse P (in kg m/s) should be imparted to the stand so that a real image of the object is formed on the screen four times during a complete oscillation. All these are separated by same time interval. (Neglect width of the stand.)



6. Electric field in a region is given by $E = 3y\hat{i} + 3x\hat{j} + 5\hat{k}$. Find out total charge enclosed in a cube of side '10cm' and sides along x , y & z axis as shown :-



Space for Rough Work

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. In a hypothetical reaction, $A(aq) \rightleftharpoons 2B(aq) + C(aq)$ (1st order decomposition). 'A' is optically active (dextro-rotatory) while 'B' and 'C' are optically inactive but 'B' takes part in a titration reaction (fast reaction) with H_2O_2 . Hence, the progress of reaction can be monitored by measuring rotation of plane polarised light or by measuring volume of H_2O_2 consumed in titration. In an experiment the optical rotation was found to be $\theta = 40^\circ$ at $t = 20$ min and $\theta = 10^\circ$ at $t = 50$ min. from start of the reaction. If the progress would have been monitored by titration method, volume of H_2O_2 consumed at $t = 15$ min. (from start) is 40 ml then volume of H_2O_2 consumed at $t = 60$ min will be :

(A) 60 ml (B) 75 ml (C) 52.5 ml (D) 90 ml

2. Virial equation of state can be written in terms of molar volume (V_m) or in terms of pressure as

$$Z = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \frac{D}{V_m^3} + \dots \quad (1)$$

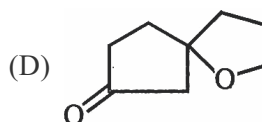
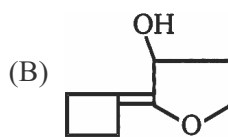
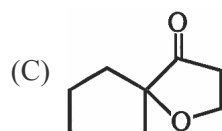
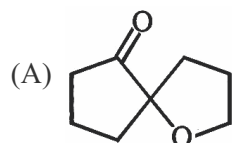
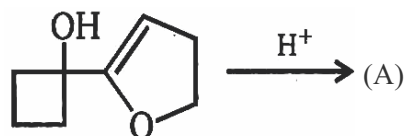
$$\text{or } Z = 1 + B'p + C'P^2 + \dots \quad (2)$$

Find C' in terms of B and C at temperature T.

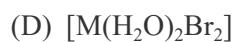
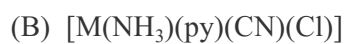
(A) $C' = \frac{C - B^2}{(RT)^2}$ (B) $C' = \frac{C - B}{(RT)^2}$ (C) $C' = \frac{C}{RT}$ (D) $C' = \frac{C}{(RT)^2}$

Space for Rough Work

3. Identify the major product :



4. Which of the following complexes has same number of stereoisomeric structures with any geometry (tetrahedral or square planar) ?



Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
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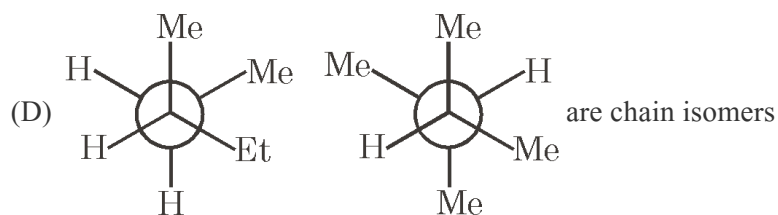
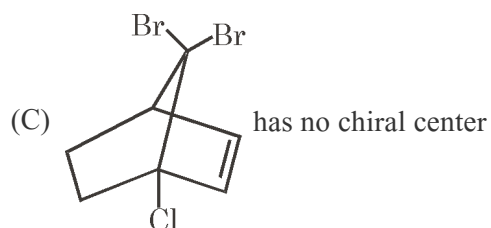
Negative Marks : -1 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

-
5. One gram of commercial AgNO_3 sample is dissolved in 50 ml of water. If is treated with 50 ml of a KI solution. The silver iodide thus precipitated is filtered off. Excess of KI in the filtrate is titrated with (M/10) KIO_3 solution in presence of 6 M HCl till all I^- ions are converted into ICl . It requires 50 mL of (M/10) KIO_3 solution, 20 mL of the same stock solution of KI requires 30 mL of (M/10) KIO_3 under similar conditions.
- (A) The % of AgNO_3 in the sample = 50%
- (B) The % of AgNO_3 in the sample = 85%
- (C) Mass of AgNO_3 in the sample = 0.5 g
- (D) Mass of AgNO_3 in the sample = 0.85 g
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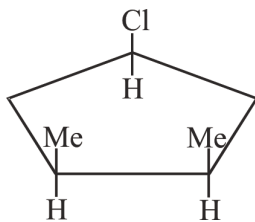
Space for Rough Work

6. In an FCC unit cell a cube is formed by joining the centers of all the tetrahedral voids to generates a new cube. Then the new cube would contain voids as :
- (A) 1 full tetrahedral void
 (B) 1 full octahedral void
 (C) 8 full tetrahedral void and 1 full octahedral void
 (D) 1 full octahedral void only
7. Which of following is/are correct :
- (A) CO_2 , Na^+ , AlCl_3 all are electrophiles.
 (B) $\text{F}^- < \text{Cl}^- < \text{Br}^- < \text{I}^-$: nucleophilicity order (in aqueous medium)

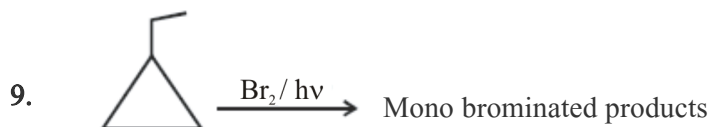


Space for Rough Work

8. Correct statement(s) about this compound :



- (A) It shows geometrical isomerism
 (B) It has centre of symmetry
 (C) It has plane of symmetry
 (D) It shows optical isomerism



- (A) Total 8 monobrominated product are form
 (B) Total 5 monobrominated product are form excluding stereoisomer
 (C) 6 monobrominated are optically active
 (D) 4 enantiomeric pair are form

Space for Rough Work

10. Which of the following statement(s) is/are **INCORRECT**?
- (A) $[\text{Pt}(\text{PPh}_3)_4]$ is diamagnetic
(B) $[\text{Pt}(\text{PPh}_3)_4]$ is dsp^2 hybridised
(C) XeO_4 is a square planar
(D) XeO_4 and $[\text{Pt}(\text{PPh}_3)_4]$ have same structure
11. Which of the following minerals do/does not contain iron ?
- (A) Rutile (B) Ilmenite
(C) Cerrusite (D) Pyrolusite
12. Compound $[\text{Co}(\text{gly})_2(\text{CN})(\text{Br})]^+$ ion has :
- (A) only one optically inactive isomer
(B) two linkage isomers
(C) six geometrical isomers
(D) eleven stereoisomers

Space for Rough Work

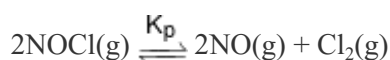
SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -0.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

1. Gaseous nitrosyl chloride and nitrogen are taken in a flask sealed and heated to some temperature where the total pressure would have been 1 bar if the following equilibrium had not been established.

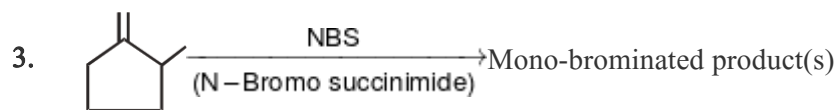


But the actual pressure was found to be 1.2 bar due to above equilibrium. Now into the above equilibrium mixture, some Cl_2 gas was introduced at constant V & T so that the total pressure would have been 8.3 bar if no further reaction had occurred but the actual pressure was found to be 8.2 bar. find the value of K_p .

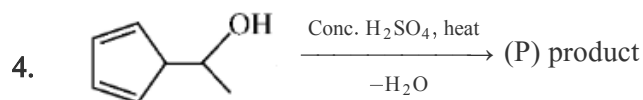
Space for Rough Work

2. At 300K, 40 ml of $O_3(g)$ dissolves in 100 g of water at 1.0 atm. What mass of ozone (in gram) dissolved in 1600 g of water at a pressure of 4.0 atm at 300 K ?

[Report your answer nearest integer value.]

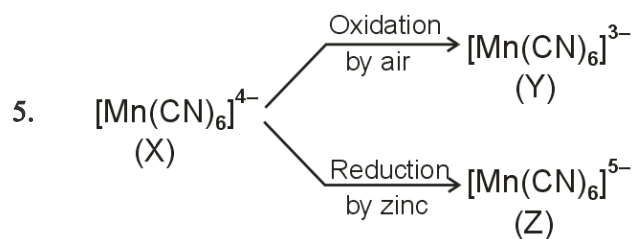


Find number of fraction obtained after fractional distillation of mono-brominated products



Degree of unsaturation in P

Space for Rough Work



if

the Crystal Field Stabilisation Energy of X is $= a \cdot \Delta_o$

the Crystal Field Stabilisation Energy of Y is $= b \cdot \Delta_o$

the Crystal Field Stabilisation Energy of Z is $= c \cdot \Delta_o$

(Neglect pairing energy)

then the value of $(a-b)/c$ is

6. A sample of ore contains

- (a) iron oxide
- (b) hydrated alumina
- (c) SiO_2
- (d) TiO_2
- (e) CuO
- (f) MgO
- (g) NiO
- (h) ZnO

Number of components soluble in excess NaOH_{aq} .

Space for Rough Work

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. If $f : [-2\pi, 2\pi] \rightarrow \mathbb{R}$ is given by $f(x) = \frac{x \cos x}{\left[\frac{x}{5\pi}\right] + \frac{1}{2}}$ where $[x]$ denotes greatest integer function then $f(x)$
 - (A) is an even function
 - (B) is an odd function
 - (C) is neither even nor odd
 - (D) satisfies $f(x) - f(-x) = 2$

2. The value of $\lim_{x \rightarrow 1} \left(\frac{2018}{1 - x^{2018}} - \frac{2019}{1 - x^{2019}} \right)$ is
 - (A) 0
 - (B) $-\frac{1}{2}$
 - (C) $\frac{1}{2}$
 - (D) does not exist

Space for Rough Work

3. A circle is drawn circumscribing the square with vertices A, B, C, D (in order) represented by $|x + y| + |x - y| = 2$. If tangents at A and B to this circle meets at P, then $PA^2 + PC^2$ is equal to
- (A) 4
(B) 8
(C) 12
(D) 20
4. In triangle ABC, vertex A is (1,1) and internal angle bisector $x - y = 2$ of angle B meets the perpendicular bisector of side AC at (4,2). If equation of side AC is $2x + y = 3$ and slope of side BC is m, then $|4m|$ is equal to
- (A) 4
(B) 3
(C) 6
(D) 7

Space for Rough Work

SECTION-I (ii) : (Maximum Marks: 32)

- This section contains **EIGHT (08)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -1 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -1 marks.

-
5. If the number of solutions of equations $\sin^{-1}x + |x| = 1$, $\tan^{-1}x + |x| = 1$ and $\sec^{-1}x + |x| = 1$ are n_1 , n_2 and n_3 respectively, then
- (A) $n_1 = 1$ (B) $n_2 = 1$ (C) $n_2 = 2$ (D) $n_3 = 1$
-

Space for Rough Work

6. A circle is tangent to line $3x - 4y + 1 = 0$ at $(1, 1)$ and is normal to line $2x + y - 5 = 0$. Choose correct option(s)?
- (A) Length of tangent to circle from $(-1, 1)$ is 4 units
- (B) Length of perpendicular from centre of circle to x-axis is 3 units.
- (C) x-intercept of director circle of given circle is $\sqrt{82}$
- (D) x-intercept of director circle of given circle is $\sqrt{41}$
7. Pairs of lines $\sqrt{3}(x-2)^2 - 4(x-2)(y-2) + \sqrt{3}(y-2)^2 = 0$ and $\sqrt{3}x^2 - Kxy + \sqrt{3}y^2 = 0$; $K \in [2\sqrt{3}, 4]$ intersect to form a quadrilateral of area A_K .
- (A) Range of $A_K \in [1, \sqrt{3}]$
- (B) Quadrilateral is rhombus $\forall K \in [2\sqrt{3}, 4]$
- (C) Range of $A_K \in [0, 8 - 4\sqrt{3}]$
- (D) Quadrilateral is kite $\forall K \in [2\sqrt{3}, 4]$
8. Let $S = \{(x, y) \mid x^2 + y^2 \geq 4\}$ and $T = \{(x, y) \mid x^2 - y^2 \leq 0\}$. Area of the region given by the set $T - S$ (in sq. units) is
- (A) π (B) 2π (C) 3π (D) 4π

Space for Rough Work

9. Let two sides AB and BC of a ΔABC be represented by $x - 7y + 3 = 0$ and $x + y + 3 = 0$ respectively. Incentre I $(-3 - 3a, a)$, $a \in \mathbb{R}$ of the triangle lies on the altitude $x - y - 1 = 0$ through A then
- (A) The equation of a line representing side AC is $7x - y = 11$
- (B) The equation of a line representing side AC is $7x + y = 1$
- (C) The area of ΔABC is $\frac{32}{3}$ sq. unit
- (D) The area of ΔABC is $\frac{32}{5}$ sq. unit
10. Let $f(x) = x^{\frac{1}{\{x\}}}$ (where $\{x\}$ denotes fractional part of x) then
- (A) $\lim_{x \rightarrow 0^+} f(2013^x) = 2013$
- (B) $\lim_{x \rightarrow 0^-} f(2013^x) = 1$
- (C) $\lim_{x \rightarrow 0^+} f(2013^x) = e$
- (D) $\lim_{x \rightarrow 0} f(2013^x)$ does not exist

Space for Rough Work

11. Which of the following statements is/are always true ?

(A) number of roots of equation $\sum_{r=1}^{2017} \frac{r}{x-r} = 0$ is 2016

(B) number of roots of equation $\sum_{r=1}^{2017} \frac{r}{x-r} = 0$ is 2017

(C) if $f(x)$ is continuous in $[0,2]$ such that $f(0) = f(2)$, then there exists atleast one $x \in [0,1]$ such that $f(x) = f(x+1)$.

(D) if $f(x)$ is continuous in $[0,2]$ such that $f(0) = f(2)$, then there exists atleast one $x \in (0,1)$ such that $f(x) = f(x+1)$.

12. If a derivable function $y = f(x)$ satisfies the condition $f(x) + f(y) = f(x)f(y) + f(xy) \forall x, y \in \mathbb{R}$ where $f(1) = 0$ and $f'(1) = -2$, then the number of solution(s) of the equation $f(x) = 2$ is less than or equal to

(A) 4

(B) 3

(C) 2

(D) 1

Space for Rough Work

SECTION-II : (Maximum Marks: 18)

- This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value to **Two** decimal places; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30, if answer is 11.36777..... then both 11.36 and 11.37 will be correct)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If ONLY the correct numerical value is entered.

Zero Marks : 0 In all other cases.

-
1. Number of points where $f(x) = |(2^x - 1).(3^x - 3)x.(x - 1).(x - 2)|$ is not differentiable, is
 2. Let $P(x) = x^{10} + ax^8 + bx^6 + cx^4 + dx^2 - 1$ be polynomial with real coefficients. If $P(1) = 1$, $P(2) = -1$ then the minimum number of real zeroes of $P(x)$ is
 3. If the sum of n terms of the series $\tan^{-1} 3 + \tan^{-1} \frac{9}{2} + \tan^{-1} \frac{33}{4} + \tan^{-1} \frac{129}{8} + \dots$ is $(2n + 1)k\pi - \tan^{-1} 2^n$, then k is
-

Space for Rough Work

4. $f(x)$ is polynomial which satisfies $|f(x)| \leq |e^x - 1| \forall x \in \mathbb{R}$ and $A = \{x : x \text{ is all possible values of } f'(0)\}$, then number of integer in set A is
5. Given two circle $x^2 + y^2 + 3\sqrt{2}(x + y) = 0$ and $x^2 + y^2 + 5\sqrt{2}(x + y) = 0$, the radius of third circle which is tangent to the two given circles and to their common diameter is
6. If $f : \mathbb{R} - \{0,1\} \rightarrow \mathbb{R}$ be a function satisfying $f(x) + f\left(1 - \frac{1}{x}\right) = \tan^{-1}x$ and $N = f\left(\frac{1}{2}\right)$, then the value of $\left(\frac{8N}{\pi}\right)$ is
-

Space for Rough Work

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work